CS & IT ENGINEERING

Data Structure & Programming

Tree

DPP- 01 Discussion Notes

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[NAT]

#Q. The number of unlabeled binary trees possible with four nodes is ___14.___.

Catalan No. $C_n = \frac{1}{n+1} {}^{2n}C_n$

$$C_4 = \frac{1}{5} {}^{8}C_4$$

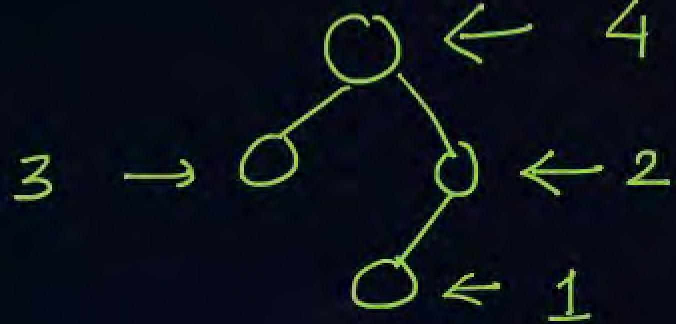
$$= \frac{1}{5} \frac{\lfloor 8}{\lfloor 4 \lfloor 4}$$

$$= \frac{1}{\cancel{5}} \frac{\cancel{8} \times 7 \times \overset{2}{\cancel{6}} \times \cancel{5}}{\cancel{4} \times \cancel{3} \times \cancel{2} \times 1} = 14$$

[NAT]

#Q. The number of labelled binary trees possible with the nodes-10, 30, 25, 40 is <u>336</u>.

No. of unlabelled trees = 14



No. of ways we ~~cant~~ can label

$= 4!$

No. of labelled trees $= 14 \times 4!$

$= 14 \times 24 = \underline{336}$

[NAT]

#Q. The number of binary search trees possible with thenodes-10, 30, 25, 40 is __14__.

No. of Binay Search tree

One Labelled tree can filled only oneway Hence

BST = # of unlabelled tres

= 14

[MCQ]

Root Left Right

Left

#Q. The pre-order traversal of a binary search tree is given as-
7, $\boxed{3, 2, 1, 5, 4, 6,}$ 8, 10, 9, 11
The post-order traversal of the above binary tree is-

A 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11

B 1, 2, 4, 6, 5, 3, 9, 11, 10, 8, 7

C 1, 2, 4, 5, 6, 3, 9, 10, 11, 8, 7

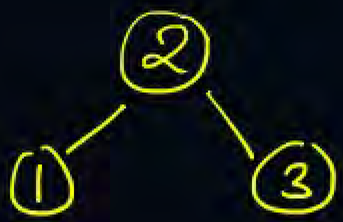
D 11, 9, 10, 8, 6, 4, 5, 1, 2, 3, 7

only preorder

postorder : 1, 2, 4, 6, 5, 3

Answer B

[MCQ]



2, 1, 3

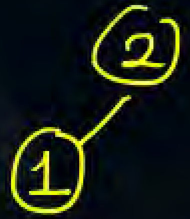
1 2 3

#Q. Consider the following two statements:

Statement P: The last elements in the pre-order and in-order traversal of a binary search tree are always same. Incorrect

Statement Q: The last elements in the pre-order and in-order traversal of a binary tree are always same. Incorrect

Which of the following tree is/are CORRECT?



preorder

2 1

Inorder

1, 2

A. Both P and Q only

B. Neither P nor Q ✓

C. Q only

D. P only

BST – Binay Tree

[MCQ]

Linked List

tree Node

#Q. Consider the following function:

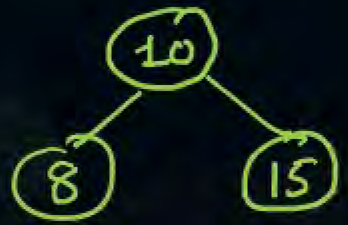
```
struct treenode{
struct treenode *left;
int data;
struct treenode *right;
};
int func (struct treenode *t){
if(t==NULL) return 1;
else if(t->left==NULL && t->right==NULL)
return 1;
else if
((t → left → data < t->data) && (t → right → data > t->data))
```

treenode pointer is Input.

(1) if t is NULL

zero

1 element



Continue...

Recursion

(BST)

Tree structure

Recursive

```
return func(t->left) && func(t->right);
else
return 0;
}
```

Logical operator && operator

Assume t contains the address of the root node of a tree. The function-

A. Returns 1 if the given tree is a Binary Search Tree.

Answer is A

B. Returns 0 if the given tree is a complete binary tree.

C. Returns 0 if the given tree is a Binary Search Tree.

D. Returns 1 if the given tree is a complete binary tree.

[MCQ]

#Q. Consider the following function:

```
struct treenode{
struct treenode *left;
int data;
struct treenode *right;
};
struct treenode * f(struct treenode *t, int x){
if(t==NULL) return NULL;
elseif(x==t->data) return _____a_____;
else if (x<t->data) return _____b_____;
else return _____c_____;
}
```

treenode structure

13

15

Data found t

f(t→Left, x)

f(t→right, x)

Continue...

Searching algorithm Root Node

[BST]

Assume t contains the address of the root node of a binary search tree. The function finds an element x in the BST and returns the address of the node if found.

Which of the following statement(s) is/are CORRECT?

Address of Node

A a: NULL ; b: f(t->left, x) ; c: f(t->right, x)

B a: t ; b: f(t->right, x) ; c: f(t->left, x)

B C a: NULL ; b: f(t->right, x) ; c: f(t->left, x)

Answer- D

C D a: t ; b: f(t->left, x) ; c: f(t->right, x) ✓

THANK - YOU